32V 500KHz 5A Fast-PWM Synchronous Step-Down Converter

DESCRIPTION

The MT8905 is a fully integrated high efficiency synchronous step-down converter which requires minimum number of external components. It offers very compact solution with up to 5A continuous output current over a wide input range.

The MT8905 employs proprietary Constant On-Time (COT) control scheme providing superior transient response and maintaining constant switching frequency under the continuous conduction mode operation. The external ramp compensation network allows stable operation with ultra-low equivalent series resistance (ESR) output ceramic capacitors. An error amplifier in the control loop provides excellent line and load regulation.

The MT8905 integrates extensive protection functions include: UVLO, OCP, OVP and thermal shutdown. Input under- voltage lockout is internally set as 3.8V.

The unique ultrasonic pulse-skipping mode (MT3905U) maintains the switching frequency above 25KHz, which eliminates noise in audio applications. Other features include pulse skipping mode (MT8905N), which maximizes efficiency in light-load applications.

The converter is available in a small 8pin SOP8_EP and DFN4x3_14L package.

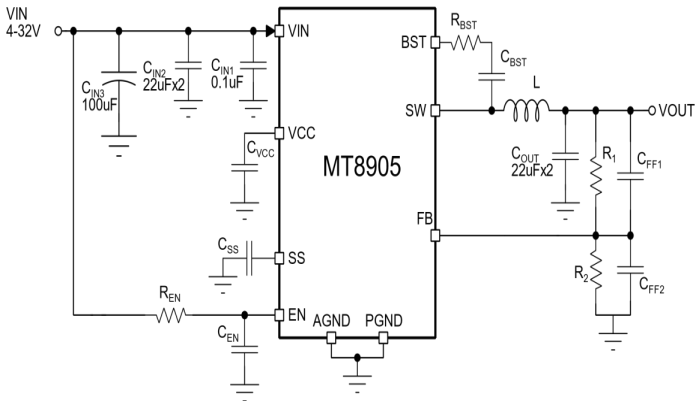
FEATURES

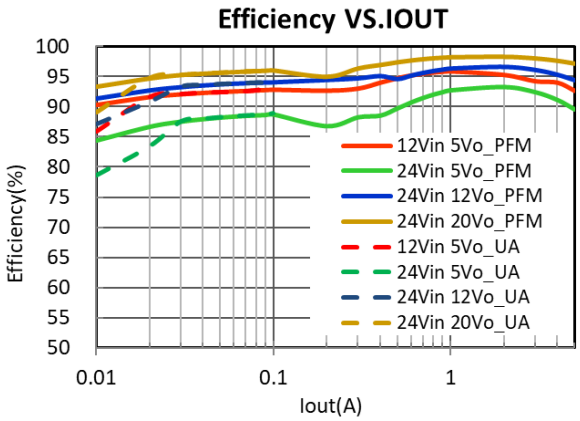
- 4V to 32V Input Voltage Range with Surge Up to 36V
- +/-2% 0.8V Feedback Voltage Accuracy
- 5A continuous output current
- Support 100% duty cycle Low Dropout Operation
- Stable operation with output low ESR ceramic capacitors
- Fast PWM Constant On Time (COT) control scheme with superior transient performance
- 500KHz Switching frequency with 2 Type Light Load Mode Control Schemes: MT8905U Ultrasonic Pulse Skip Mode (25KHz Min); MT8905N Pulse Skip Mode
- Integrated 40mΩ and 40mΩ High Side and Low Side Switches
- Accurate EN Threshold for the External Programmable VIN UVLO
- Low quiescent current
- Thermal Shutdown with Auto recovery.
- Available in SOP8_EP and DFN4x3_14L Package

APPLICATIONS

- Laptop Computer
- Tablet PC
- Networking Systems
- Personal Video Recorders
- Flat Panel Television and Monitors
- Distributed Power Systems

TYPICAL APPLICATIONS



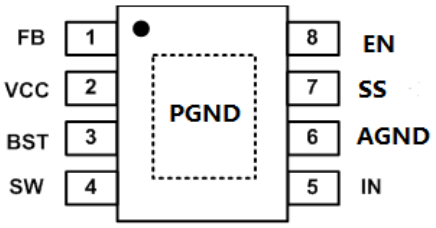


ORDERING INFORMATION

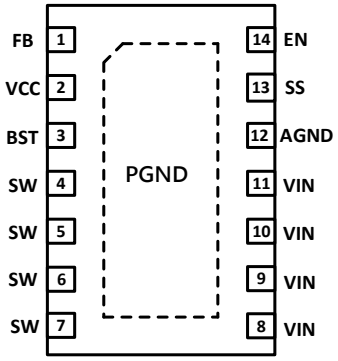
Part No.	Marking	Temp. Range	Remark	Package	MOQ
MT8905NSPR	MT8905 YWWXX	-40°C ~85°C	PFM mode	SOP8_EP	2500/Tape & Reel
MT8905USPR	MT8905U YWWXX	-40°C ~85°C	UA mode	SOP8_EP	2500/Tape & Reel
MT8905NDDR	MT8905 YWWXX	-40°C ~85°C	PFM mode	DFN4x3_14L	5000/Tape & Reel
MT8905UDDR	MT8905U YWWXX	-40°C ~85°C	UA mode	DFN4x3_14L	5000/Tape & Reel

Note: Y: Year, WW: Week, XX: Control Code

PIN CONFIGURATION



SOP8_EP



DFN4x3_14L

PIN DESCRIPTION

SOP8_EP Pin No	DFN4x3_14L Pin No	SYMBOL	DESCRIPTION
1	1	FB	Feedback Input. Connect FB to the center of the external resistor divider from the output to the AGND to set the output voltage.
2	2	VCC	Internal 5V LDO output. The driver and control circuits are powered from this voltage. Decouple with a minimum 1μF ceramic capacitor to PGND as close to the pin as possible.
3	3	BST	High-Side Driver Bootstrap Supply. Connect a 0.1uF capacitor between SW and BST for proper operation.
4	4,5,6,7	SW	Output pin of internal power switches. Connect this pin to the inductor and bootstrap capacitor.
5	8,9,10,11	IN/VIN	Supply Voltage. The IN pin supplies power for internal MOSFET and regulator. The MT8905 operates from a +4V to +32V input rail. Bypass IN to PGND with 22μF or greater low ESR ceramic capacitor.
6	12	AGND	System Analog Ground.

SOP8_EP Pin No	DFN4x3_14L Pin No	SYMBOL	DESCRIPTION
7	13	SS	Output Soft-Start Pin. This pin allows user control of output voltage ramp rate during start-up. An internal 10μA pull-up current from VCC on this pin allows a capacitor to program output voltage slew rate. This pin is pulled to ground with an internal MOSFET during shutdown and fault conditions.
8	14	EN	The device shuts down when this pin is low and active when the pin is high. The hysteretic threshold voltage is 1.3V on the rising edge and 1.2V on the falling edge. An external resistor divider from VIN can be used to set a VIN threshold below which the device will shut down. Connect the EN to VIN through a 100kΩ resistor for automatic startup.
EP	EP	PGND	Exposed Pad is connected to the low side MOSFET Power Ground. Connect EP to a large-area contiguous copper ground plane for effective power dissipation.

FUNCTIONAL BLOCK DIAGRAM

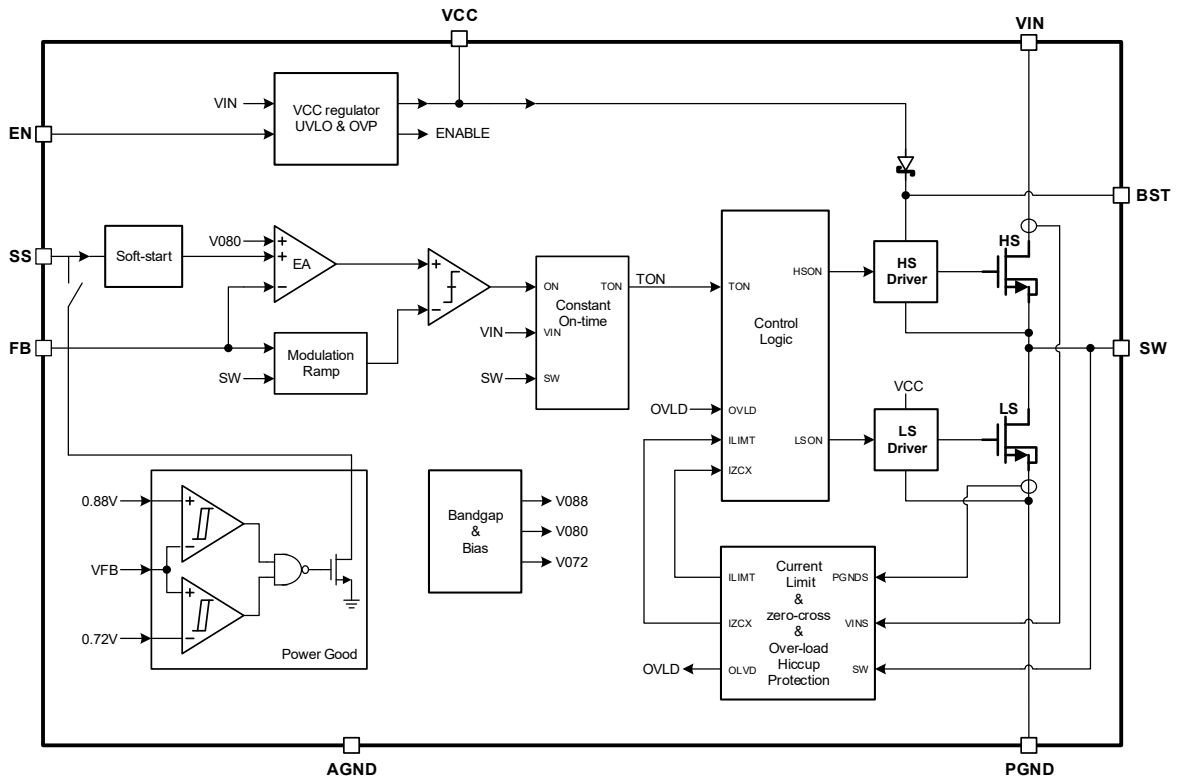


Figure 1—Functional Block Diagram

Absolute Maximum Ratings (Reference to AGND) (1)

Supply Voltage IN to GND..... +36V
EN to GND..... +36V
SW to GND..... -0.3V to V_{IN}+0.3V
SW Surge (30ns) to GND…….....-3V to V_{IN}+3V
PGND to GND ..... -0.3V to +0.3V
BST to SW..... –0.3V to +6V
All Other Pins to GND..... –0.3V to +6V
Lead Temperature ..... +260°C
Junction Temperature ……..... +150°C
Storage Temperature .....-65°C to +150°C

Recommend Operating Conditions (2)

Supply Voltage VIN .....+4V to +32V
Output Voltage VOUT ..... 0.8V to V_{IN}
Operating Junction Temp. (TJ)……. -40°C to +125°C

Thermal Information (3)(4)

Maximum Power Dissipation(T_A=+25°C) .
SOP8_EP.....…….....…….....…….....… 2.77W
DFN4x3_14L……..2.57W
Thermal Resistance (θ_{JA}) SOP8_EP .....…..45°C/W
Thermal Resistance (θ_{JC}) SOP8_EP..... 38°C/W
Thermal Resistance (θ_{JA}) DFN4x3_14L..... 48.5°C/W
Thermal Resistance (θ_{JC}) DFN4x3_14L.....…….28°C/W

Note(1): Stress exceeding those listed “Absolute Maximum Ratings” may damage the device.
Note(2): The device is not guaranteed to function outside of the recommended operating conditions.
Note(3): Measured on JESD51-7, 4-Layer PCB.
Note(4):The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by $P_{D_MAX}= (T_{J_MAX}-T_A)/\theta_{JA}$. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

ELECTRICAL CHARACTERISTICS

T_A = +25°C, V_{IN}=12V and V_{EN}=2V, unless otherwise noted. Typical values are at V_{IN} = 12V, V_{EN} = 2V and V_{OUT} = 5V.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input Voltage Range V_{IN}	V_{IN}		4		32	V
Shutdown Current	$I_{SHUTDOWN}$	V_{EN} = 0V, V_{IN} = 12V		6		μA
Input Under Voltage Lockout Threshold	V_{UVLO}	V_{IN} rising		3.80	4.05	V
Input Under Voltage Lockout Hysteresis	V_{UVHYS}			460		mV
Quiescent Current	I_Q	V_{FB} = 0.84V (PFM mod)	90	125	160	μA
		V_{FB} = 0.84V (UA mod)	160	210	260	
Feedback Regulation Voltage	V_{FB}		784	800	816	mV
Feedback Current	I_{FB}	V_{FB}=0.8V		1		nA
Internal Soft-start Time	T_{SS}		2.5	2.9	3.3	msec
Switching Frequency	F_{SW}	I_{OUT} = 1A		500		KHz
		Ultrasonic Mode MT8905U I_{OUT} = 0A	25	30		
Minimum Off Time (5)	T_{OFF_MIN}			100		ns
Maximum Duty Cycle (6)	D_{MAX}	V_{FB} = 0.7V		100		%
HS Main Switch On Resistance RON_{HS}	HS_{RDS-ON}	MT8905NSPR/MT8905USPR		40		mΩ
		MT8905NDDR/MT8905UDDR		35		
HS Switch Leakage Current(7)	SW_{LKG}	V_{IN} =32V, V_{EN} = V_{SW} = 0V		0.1	1	μA
HS Switch Current Limit	I_{PK}		6	7.5	9.5	A
Low Side(LS) Switch Valley Current Limit	I_{VALLEY}		5	6.5	8	A
LS Switch Zero-Cross Current Limit	I_{ZX}			0		mA
LS Switch On Resistance RON_{LS}	LS_{RDS-ON}	MT8905NSPR/MT8905USPR		36		mΩ
		MT8905NDDR/MT8905UDDR		35		
LS Switch Leakage Current(7)		V_{IN} = V_{SW}= 32V, V_{EN} = 0V		0.1	1	μA
EN On Threshold	V_{EN_ON}	V_{EN} ramp up		1.3		V
EN Off Threshold	V_{EN_OFF}	V_{EN} ramp down		1.2		V
EN Internal Pull Down Resistor				1		MΩ
VCC Linear Regulator	V_{CC}	V_{FB} = 0.84V, I_{VCC}=0~30mA	5	5.3	5.5	V
Thermal Shutdown	T_{SD}			160		°C
Thermal Shutdown Hysteresis				30		°C
Output Discharging Resister	R_{DIS}			15		Ω

Note(5): Guaranteed by design, no production test.

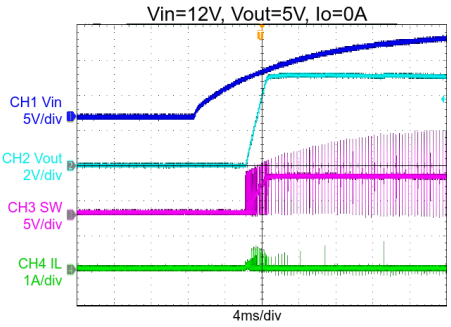
Note(6): When the input voltage approaches the output voltage, the MT8905 will extend the on-time and force the main high side switch remaining on for multiple cycles (>10 μsec). High side switch is only turned off momentarily, and low side switch is forced on shortly (typical 120ns) to refresh the BST capacitor. High side switch will resume on after the BST refresh.

Note(7): When using the discharge function, the HS/LS leakage cannot be tested at the user end.

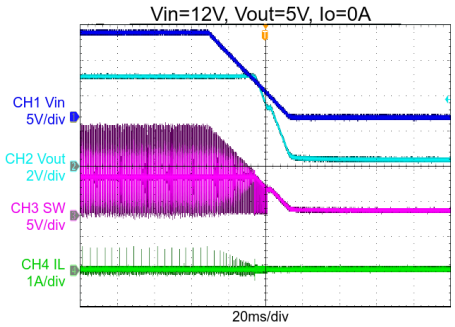
TYPICAL PERFORMANCE CHARACTERISTICS

Performance waveforms are tested on the evaluation board of the Design Example section.
VIN=12V, VOUT =5V, L=6.8μH, TJ=+25°C, unless otherwise noted.

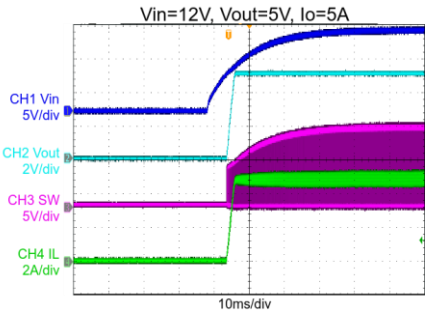
Vin Power On



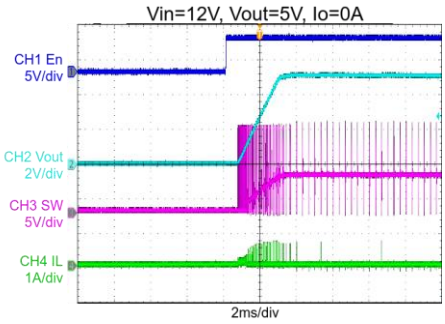
Vin Power Off



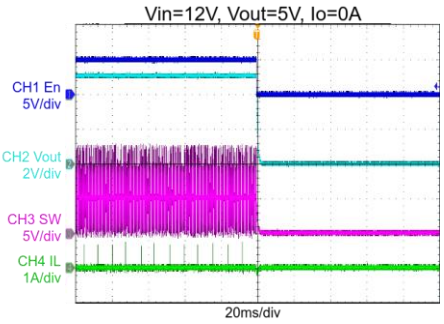
Vin Power On



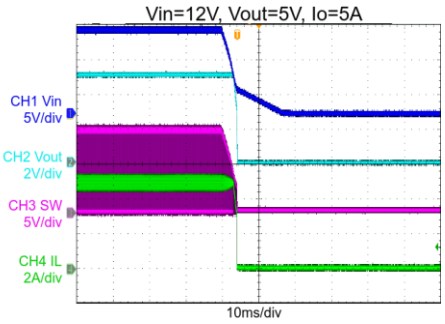
EN Power On



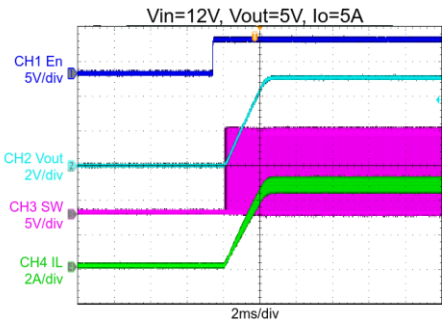
EN Power Off



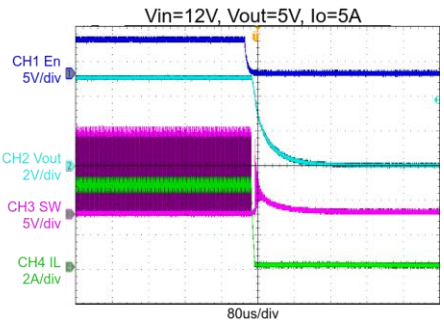
Vin Power Off



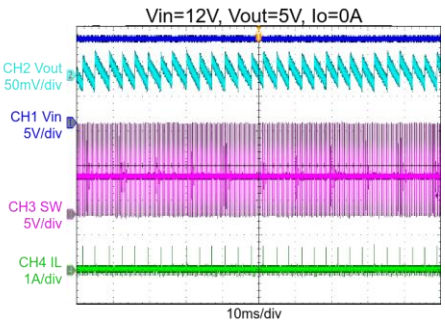
EN Power On



EN Power Off



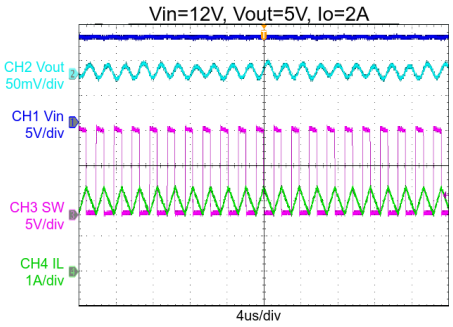
Steady State Test



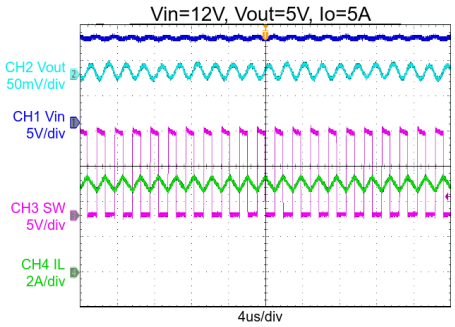
TYPICAL PERFORMANCE CHARACTERISTICS *(continued)*

Performance waveforms are tested on the evaluation board of the Design Example section.
VIN=12V, VOUT =5V, L=6.8μH, TJ=+25°C, unless otherwise noted.

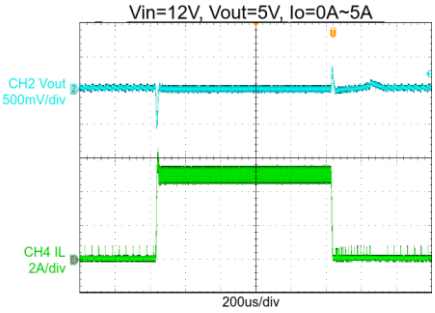
Steady State Test



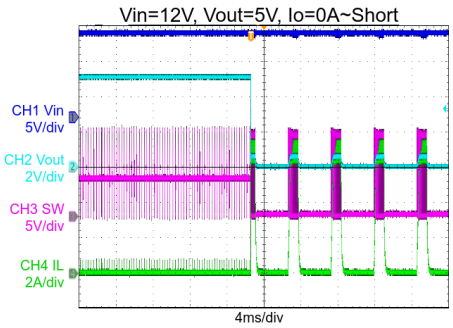
Steady State Test



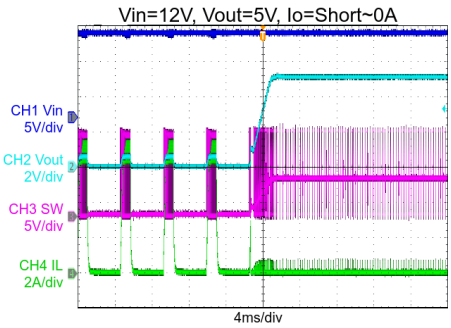
Load Transient Response



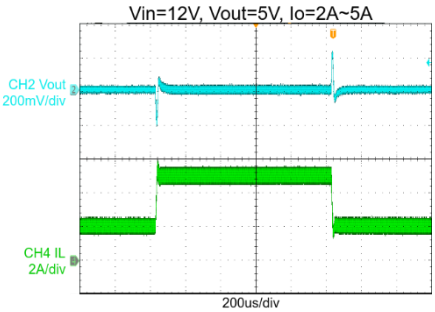
Short Protection



Short Protection



Load Transient Response



OPERATION

The MT8905 is a fully integrated synchronous step-down converter employing constant on-time (COT) control scheme to achieve superior transient performance. Its proprietary external ramp compensation offers stable operation with lower ESR ceramic output capacitors.

Constant On-Time Control

The constant on-time control (COT) operates by comparing the feedback voltage V_{FB} with the reference voltage (V_{FBREG}). When FB drops below the reference, the control circuit turns on HS switch immediately for a pre-determined period of time (on-time) to ramp up the inductor current. When this on-time times out, the LS switch is then turned on to ramp down the inductor current. The LS switch is turned off when inductor current reaches zero I_{ZX} or HS switch is turned on again for the next cycle. This operation repeats itself if FB drops below reference again.

The MT8905 uses a proprietary algorithm to calculate the on-time based on input voltage, output voltage and load current to achieve nearly constant switching frequency over entire continuous conduction load current range. The on-time can be estimated as:

$$T_{ON} = \frac{V_{OUT}}{V_{IN}} \times 2uS$$

Due to its immediate response on FB voltage droop and simplified loop compensation, The MT8905 offers superior transient response compared to traditional fixed frequency PWM control converters.

Light Load Operation

In medium and heavy load conditions, the MT8905 operates in PWM mode with a typical switching frequency of 500kHz. When load current reduces, the MT8905 naturally transitions from PWM mode to PFM mode where the pulse width remains calculated on-time, but the switching frequency reduces to accommodate the low output current. The lower the output current, the lower the switching frequency. Once the switching frequency drops to low enough, the devices enter sleep mode to cut down its quiescent current to maintain high efficiency in light load.

The critical load current at the boundary of PWM mode and PFM mode is related to the inductor ripple current, which depends on the inductor value, input voltage and output voltage. Typically this critical load current level is estimated as:

$$I_{CRIT} = \frac{1}{2}\frac{(V_{IN} - V_{OUT}) \times V_{OUT}}{L \times F_{SW} \times V_{IN}}$$

When switching frequency drops below 20KHz in very light load condition, the switching noise can be heard by the human ears. MT3905U offers a unique pulse-skipping mode with a minimum switching frequency of 25KHz. This ultrasonic pulse-skipping mode reduces audio-frequency modulation of the power supply that may occur at very-light loads. The transition to fixed-frequency PWM operation is automatic and occurs at the same point as in pulse-skipping

Mode. Ultrasonic pulse skipping occurs if no switching has taken place within the last 30μs. Low side switch turns on for a short period to induce a negative current in the inductor to discharge the output. High side turns on if the output drops below the regulation point.

100% Duty Cycle Low Dropout Operation

When input voltage approaches the output voltage, the MT8905 will extend the on-time toward the maximum on-time to satisfy the duty cycle requirement to regulate the output voltage. If the input further drops to equal or lower than the output level, the MT8905 forces the main high side (HS) switch to remain on for more than one cycle, eventually reaching 100% duty cycle. The 100% duty cycle operation allows the converter to effectively pass through the input voltage directly to output with minimum voltage drops on the HS switch and the inductor. In the low dropout operation mode, the MT8905 turns on HS switch for multiple switching cycles until it turns off HS switch momentarily and turns on low side (LS) switch (typical 120ns) to refresh the BST supply voltage. The LS switch is turned off after the BST refresh pulse, then the HS switch resumes on for multiple switching cycles which gives the effective 100% duty cycle. The refresh BST pulse is needed to charge the BST capacitor and ensure the HS switch driver circuit proper operation.

Enable

MT8905 offers an accurate enable threshold of EN pin, which is typically 1.3V rise and 1.2V fall. The MT8905 is enabled by pulling up the EN pin above 1.3V and MT8905 is disabled by pulling down the EN pin above below 1.2V.

When using the EN pin threshold voltage to program the startup input voltage level, the following equation shall be used:

$$V_{IN-START} = 1.3V \times \frac{R_{UP} + R_{DOWN}//1M}{R_{DOWN}//1M}$$

Where the 1MegΩ is the internal pull down resistor on EN pin.

When EN is pulled high, MT8905 will start up if V_{IN} is higher than UVLO threshold. When EN is pulled low, MT8905 will go into shutdown. Tie EN pin to V_{IN} if the shutdown feature is not used.

Soft Start

MT8905 has built-in soft start of 2.9 msec. During the soft start period, output voltage is ramped up linearly to the regulation level, independent of the load current and output capacitor value. SS pin allows user control of output voltage ramp rate during start-up. An internal 10μA pull-up current from VCC on this pin allows a capacitor to program output voltage slew rate. The soft start time with different capacitor is below:

$$T_{SS} = \frac{0.8V \times C_{SS}}{10uA}$$

Current Limit and Hiccup Mode

The MT8905 has built-in cycle-by-cycle current limit protection to prevent inductor current from running away in any fault conditions. The MT8905 continuously monitors the inductor valley current during its operation. Once the valley current exceeds the limit level, MT8905 will turn on LS and wait for the inductor current to drop down to a pre-determined level before the HS can be turned on again. If this current limit condition is repeated for a sustained long period of time, MT8905 will enter hiccup mode, where it stops switching for a pre-determined period of time before automatically re-try to start up again. It always starts up with soft-start to limit inrush current and avoid output overshoot.

When MT8905 enters valley current limit mode, the peak current is also limit due to the fixed on-time of the HS, and this peak current can be estimated as:

$$I_{PEAK} = I_{VALLEY} + T_{ON} \times \frac{(V_{IN} - V_{OUT})}{L}$$

APPLICATION INFORMATION

Setting the Output Voltage

The external feedback resistors divider sets the output voltage (see Typical Application Circuit on page 1). 1% of resistors are recommended to maintain output voltage accuracy. The feedback resistor R1 has some impact on the loop stability with the internal compensation capacitor. Choose a value for R1, R2 is then given by:

$$R_2 = \frac{R_1}{\frac{V_{OUT}}{0.8} - 1}$$

Inductor

The inductor is necessary to supply constant current to the output load while being driven by the switched input voltage. A larger-value inductor will result in less ripple current that will result in lower output ripple voltage. However, a larger-value inductor will have a larger physical footprint, higher series resistance, and/or lower saturation current. A good rule for determining the inductance value is to design the peak-to-peak ripple current in the inductor to be in the range of 30% to 40% of the maximum output current, and that the peak inductor current is below the maximum switch current limit. The inductance value can be calculated by:

$$L = \frac{V_{OUT}}{F_{SW} \times \Delta I_L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Where ΔIL is the peak-to-peak inductor ripple current.

To avoid overheating and poor efficiency, an inductor must be chosen with an RMS current rating that is greater than the maximum expected output load of the application. In addition, the saturation current (typically labeled ISAT) rating of the inductor must be higher than the maximum load current plus 1/2 of in inductor ripple current.

The peak inductor current can be calculated by:

$$I_{LP} = I_{OUT} + \frac{V_{OUT}}{2F_{SW} \times L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Input Capacitor

The input current to the step-down converter is discontinuous and therefore requires a capacitor to supply the AC current to the step-down converter while maintaining the DC input voltage. Ceramic capacitors are recommended for best performance and should be placed as close to the VIN pin as possible. Capacitors with X5R and X7R ceramic dielectrics are recommended because they are fairly stable with temperature fluctuations. The capacitors must also have a ripple current rating greater than the maximum input ripple current of the converter. The input ripple current can be estimated as follows:

$$I_{CIN} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})}$$

The worst-case condition occurs at V_{IN} = 2*V_{OUT}, where:

$$I_{CIN} = \frac{I_{OUT}}{2}$$

For simplification, choose the input capacitor with an RMS current rating greater than half of the maximum load current. The input capacitance value determines the input voltage ripple of the converter. If there is an input voltage ripple requirement in the system, choose the input capacitor that meets the specification. The input voltage ripples can be estimated as follows:

$$\Delta V_{IN} = \frac{I_{OUT}}{F_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

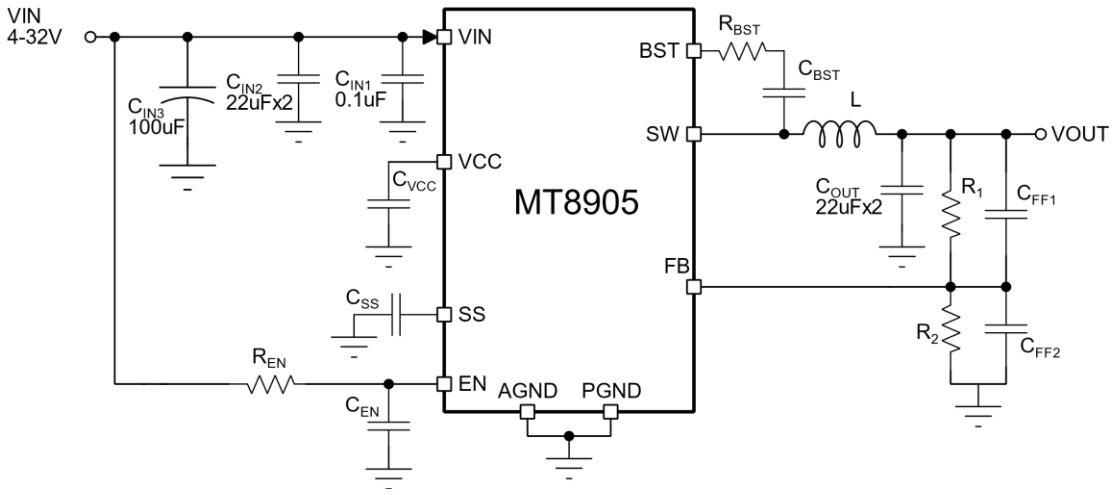
Under worst-case conditions where V_{IN} = 2*V_{OUT}:

$$\Delta V_{IN} = \frac{1}{4} \times \frac{I_{OUT}}{F_{SW} \times C_{IN}}$$

Output Capacitor

The output capacitor has two essential functions. Along with the inductor, it filters the square wave generated by the MT8905 to produce the DC output. In this role it determines the output ripple, thus low impedance at the switching frequency is important. The second function is to store energy in order to satisfy transient loads and stabilize the MT8905's control loop. X5R or X7R type ceramic capacitors have very low equivalent series resistance (ESR) and provide low output ripple and good transient response. Transient performance can be improved with a higher value output capacitor. Increasing the output capacitance will also decrease the output voltage ripple. A lower value of output capacitor can be used to save space and cost, but transient performance will suffer and may cause loop instability. When choosing a capacitor, special attention should be given to the data sheet to calculate the effective capacitance under the relevant operating conditions of voltage bias and temperature. A physically larger capacitor or one with a higher voltage rating may be required.

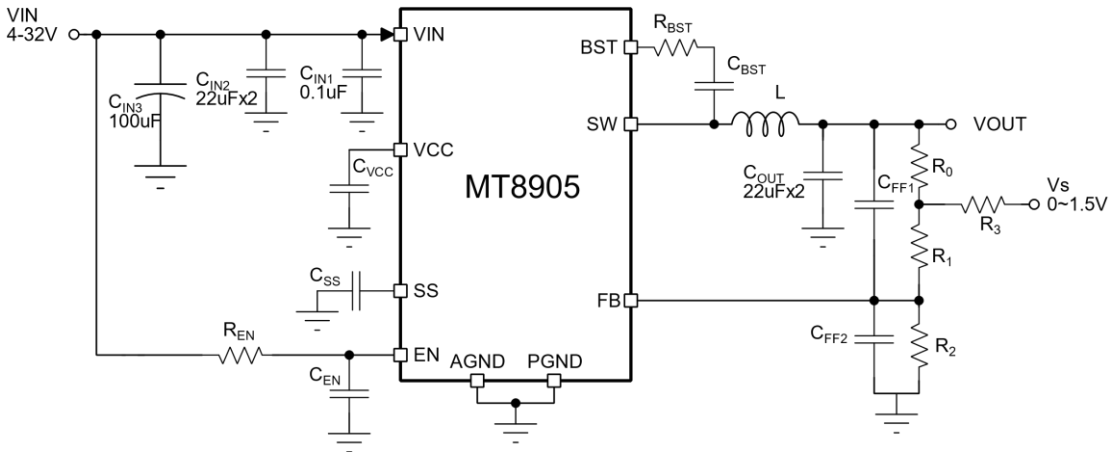
Application Schematic1: USB PD Application



EVB BOM List

Qty	Ref	Value		Description	Package
1	C_{IN3}	100μF		Electrolytic Capacitor, 50V	8x12mm
2	C_{IN2}	22μF		Ceramic Capacitor, 35V, X5R	0805
2	C_{IN1}, C_{EN}	0.1μF		Ceramic Capacitor, 50V, X5R	0603
2	C_{OUT}	22μF		Ceramic Capacitor, 35V, X5R	0805
1	C_{BST}	0.1μF		Ceramic Capacitor, 50V, X5R	0603
1	R_{BST}	0ohm		Resistor, ±1%	0603
1	C_{VCC}	1uF		Ceramic Capacitor, 50V, X5R	0603
1	L	Vout=3.3V	4.7μH	Wurth Electronics 74439384047, DCR=24.9 mΩ	SMD
		Vout=5V	6.8μH	Wurth Electronics 74439387068, DCR=16.8 mΩ	
		Vout=20V	10uH	Wurth Electronics 74439387100, DCR=22.9 mΩ	
1	R_1	Vout=3.3V	120KΩ	Resistor, ±1%	0603
		Vout=5V	120KΩ		
		Vout=20V	120KΩ		
1	R_2	Vout=3.3V	38.3KΩ	Resistor, ±1%	0603
		Vout=5V	22.6KΩ		
		Vout=20V	4.99KΩ		
1	C_{FF1}	Vout=3.3V	47pf	Ceramic Capacitor, 50V, X5R	0603
		Vout=5V	47pf		
		Vout=20V	22pf		
1	C_{FF2}	Vout=3.3V	NC	Ceramic Capacitor, 50V, X5R	0603
		Vout=5V	NC		
		Vout=20V	NC		
1	R_{EN}	0Ω~100KΩ		Resistor, ±1%	0603
1	Power IC	MT8905		Step-Down DC/DC Converter	SOP8_EP DFN4x3_14L

Application Schematic2: Type-c Application



EVB BOM List

Qty	Ref	Value	Description	Package
1	C_{IN3}	100μF	Electrolytic Capacitor, 50V	8x12mm
2	C_{IN2}	22μF	Ceramic Capacitor, 35V, X5R	0805
2	C_{IN1}, C_{EN}	0.1μF	Ceramic Capacitor, 50V, X5R	0603
2	C_{OUT}	22μF	Ceramic Capacitor, 35V, X5R	1206
1	C_{BST}	0.1μF	Ceramic Capacitor, 50V, X5R	0603
1	R_{BST}	0ohm	Resistor, ±1%	0603
1	C_{VCC}	1uF	Ceramic Capacitor, 50V, X5R	0603
1	L	10uH	Wurth Electronics 74439387100, DCR=22.9 mΩ	SMD
1	R_0	100KΩ	Resistor, ±1%	0603
1	R_1	20KΩ	Resistor, ±1%	0603
1	R_3	10KΩ	Resistor, ±1%	0603
1	R_2	22.6 KΩ	Resistor, ±1%	0603
1	C_{FF1}	22pf	Ceramic Capacitor, 50V, X5R	0603
1	C_{FF2}	NC	Ceramic Capacitor, 50V, X5R	0603
1	R_{EN}	0Ω~100KΩ	Resistor, ±1%	0603
1	Power IC	MT8905	Step-Down DC/DC Converter	SOP8_EP DFN4x3_14L

PCB Layout Recommendation

(1) The high current paths (PGND, IN, and SW) should be placed very close to the device with short, direct and wide traces.

(2) Put the input capacitors as close to the IN and PGND pins as possible.

(3) Put the decoupling capacitor as close to the VCC and GND pins as possible. Place the Cap close to GND if the distance is long. And place >3 Vias if via is required to reduce the parasitic inductance.

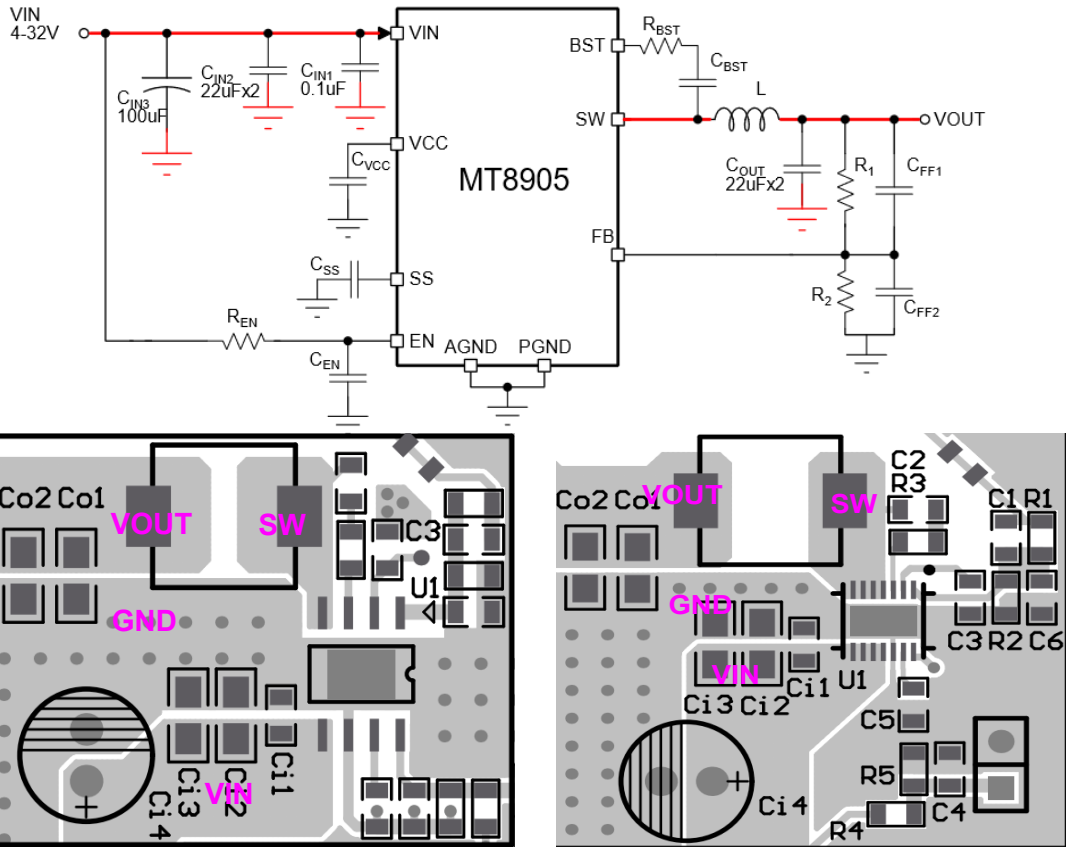
(4) Keep the switching node SW short and away from the feedback network.

(5) The external feedback resistors should be placed next to the FB pin. Make sure that there is no via on the FB trace.

(6) Keep the BST voltage path (BST, RBST, CBST and SW) as short as possible.

(7) Keep the IN and PGND pads connected with large copper and use at least two layers for IN and PGND trace to achieve better thermal performance. Also, add several Vias with 10mil_drill/18mil_copper_width close to the IN and PGND pads to help on thermal dissipation.

(8) Four-layer layout is strongly recommended to achieve better thermal performance.

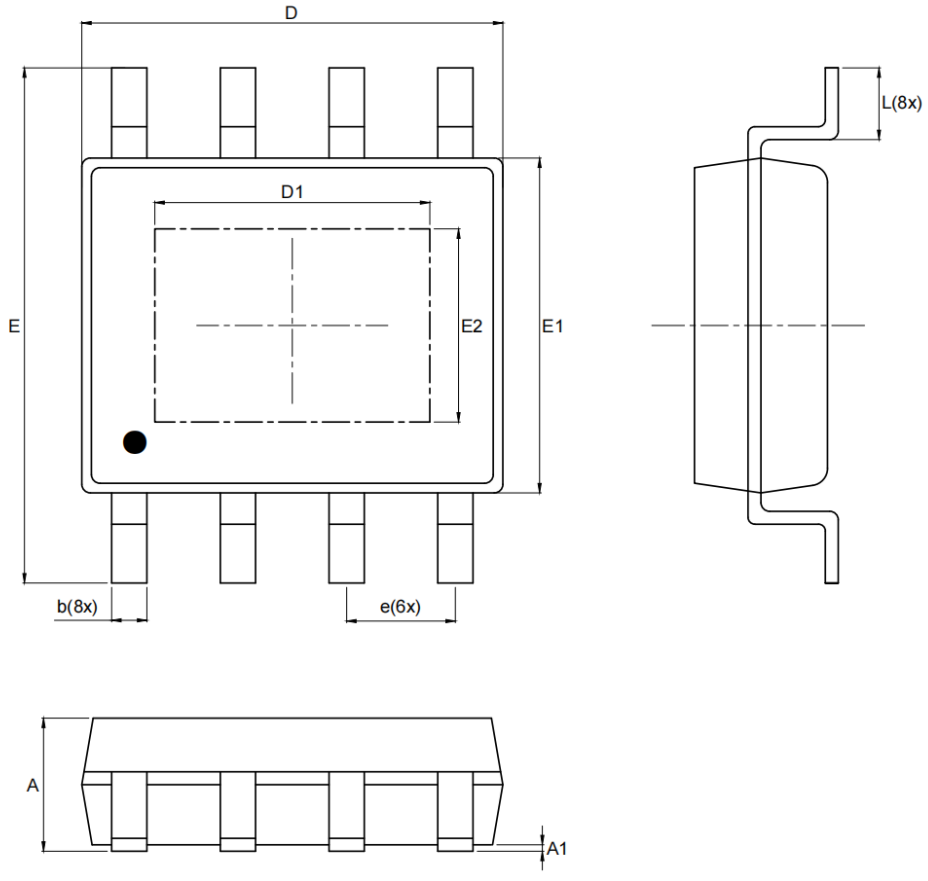


SOP8_EP DFN4x3_14L

PACKAGING INFORMATION

SOP8_EP Outline Dimensions

Unit: inches/mm

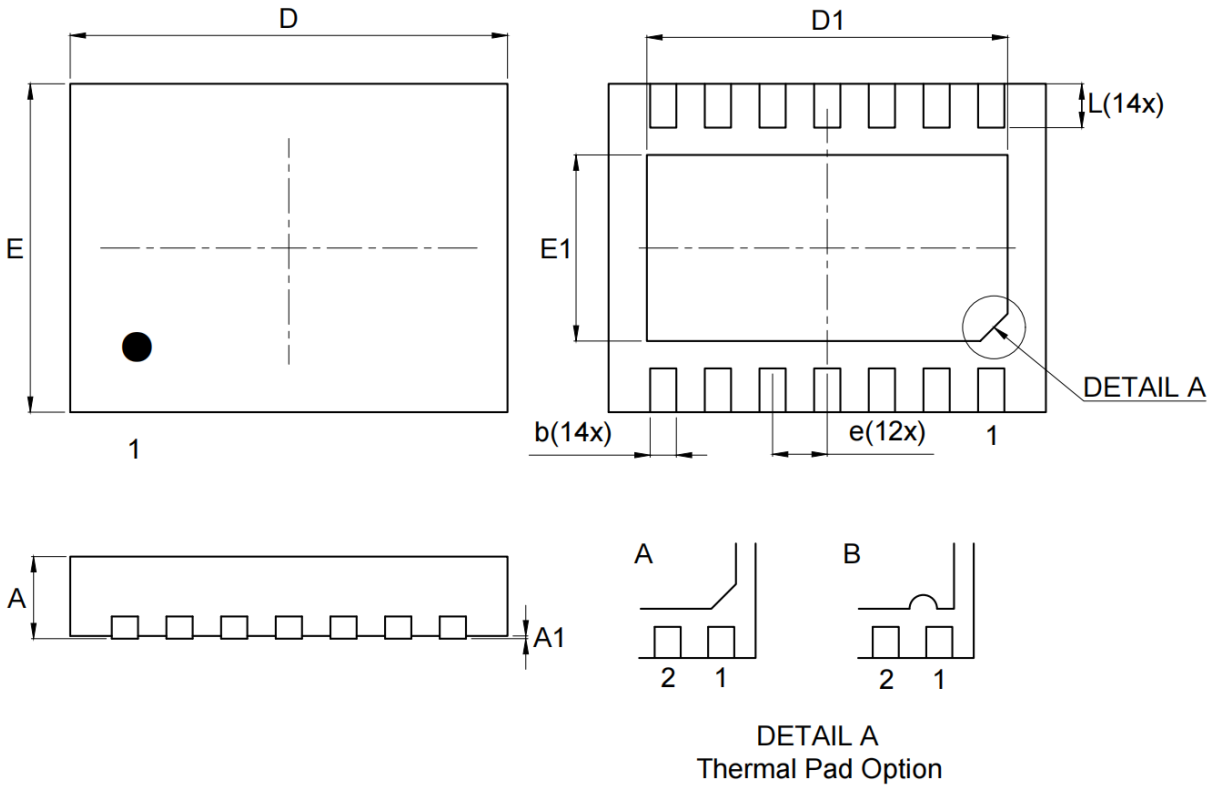


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	1.35	1.75	0.053	0.069
A1	0.00	0.15	0.000	0.006
D	4.70	5.10	0.185	0.201
E1	3.70	4.10	0.146	0.161
D1	2.90	3.50	0.114	0.138
E2	2.00	2.50	0.079	0.098
E	5.80	6.20	0.228	0.244
L	0.40	1.27	0.016	0.050
b	0.31	0.51	0.012	0.020
e	1.27 BSC		0.050 BSC	

PACKAGING INFORMATION

DFN4x3_14L Outline Dimensions

Unit: inches/mm

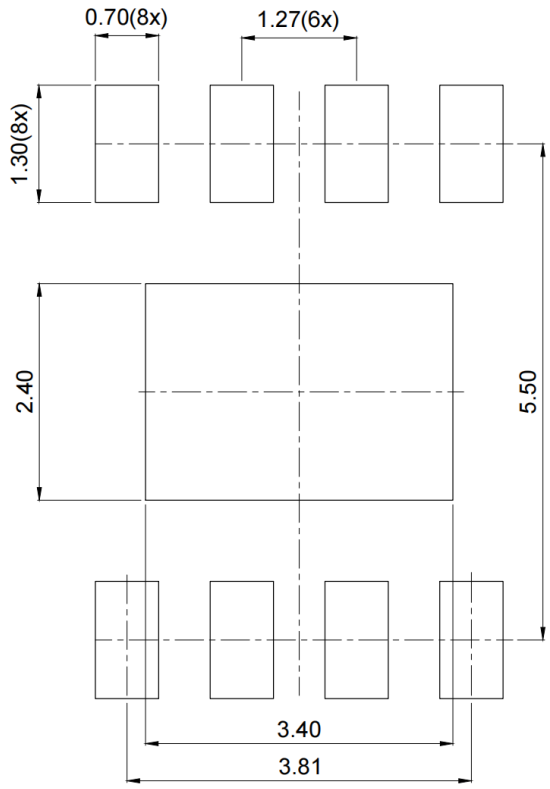


DETAIL A
Thermal Pad Option

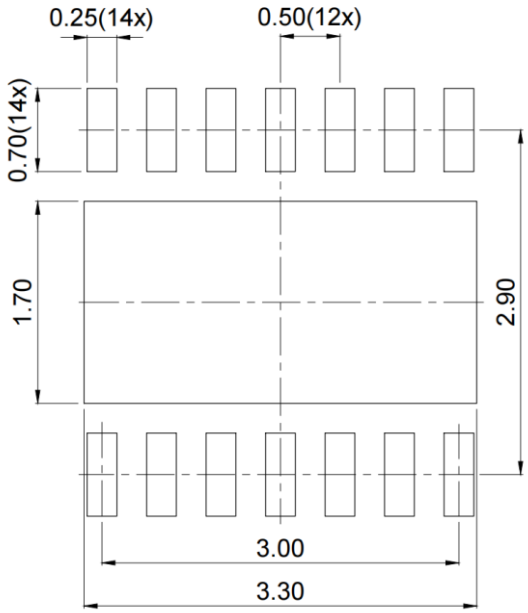
SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.70	0.80	0.028	0.031
A1	0.00	0.05	0.000	0.002
b	0.18	0.30	0.007	0.012
D	3.90	4.10	0.154	0.161
D1	3.20	3.40	0.126	0.134
E	2.90	3.10	0.114	0.122
E1	1.60	1.80	0.063	0.071
e	0.50 BSC		0.020 BSC	
L	0.30	0.50	0.012	0.020

LAND PATTERN INFORMATION

SOP8_EP



DFN4x3_14L

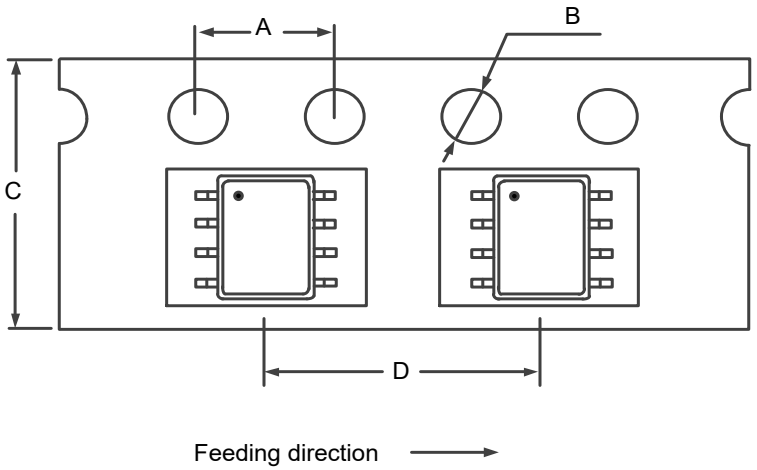


Note:

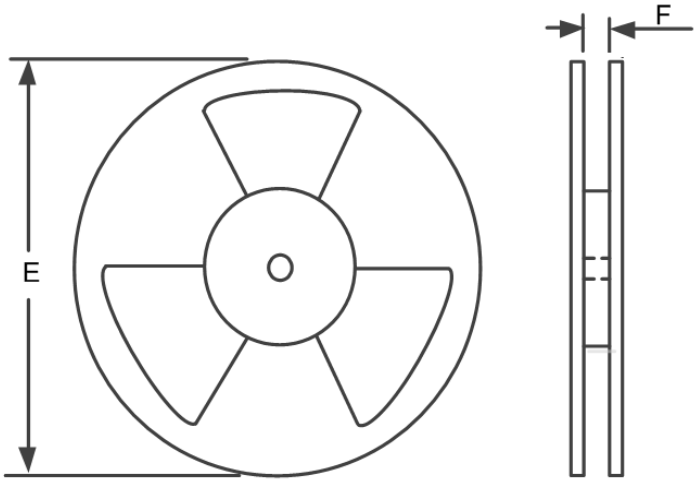
- Dimensions in mm.
- For reference only.

CARRIER TAPE & REEL DIMENSIONS

1. Orientation / Carrier Tape Information:



2. Reel Information:

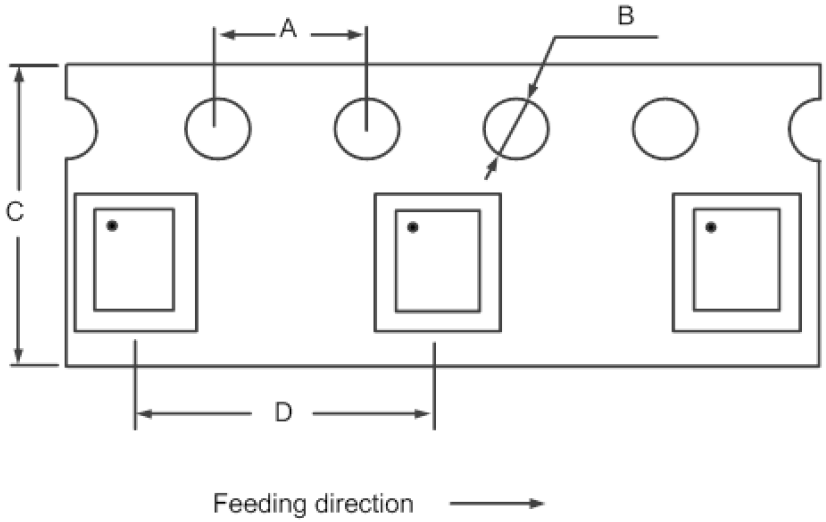


3. Dimension Details:

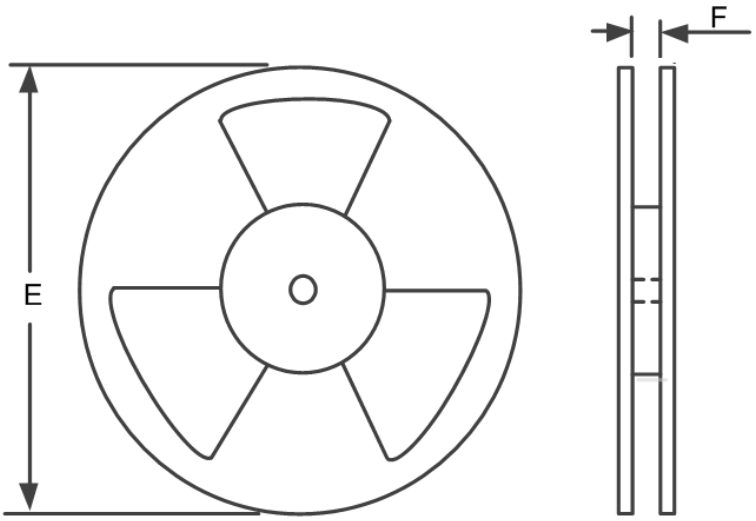
PKG Type	A	B	C	D	E	F	Q'ty/Reel
SOP8_EP	4.0 mm	1.5 mm	12.0 mm	8.0 mm	13 inches	13.0 mm	2,500

CARRIER TAPE & REEL DIMENSIONS

1. Orientation / Carrier Tape Information:



2. Reel Information:



3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
DFN4x3_14L	4.0 mm	1.5 mm	12.0 mm	8.0 mm	13 inches	13.0 mm	5,000

REFLOW PROFILE

Classification Of IR Reflow Profile

Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ts_{min} to Tp)	1~2℃/second, 3℃/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time(ts_{min} to ts ts_{max})	 150℃ 200℃ 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217℃ 60~150 seconds
Peak Temperature(Tp)	See Classification Temp intable1
Time within 5℃of actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6℃/second max.
Time 25℃ to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature(T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p) is defined as a supplier minimum and a user maximum.

Table 1. Pb-freeProcess –Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6mm	260 ℃	260 ℃	260 ℃
1.6mm–2.5mm	260 ℃	250 ℃	245 ℃
2.5mm	250 ℃	245 ℃	245 ℃

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.

